

Active Inference ~ Parr, Pezzulo, Friston ~ Chapter 2 ~ BookStream #002.0

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On to chapter two, the low road to active inference.

So all chapters begin with a short quotation,

and the quotation here reads,

my thinking is first and last and always for the sake of my doing, William James.

So even before reading the chapter,

these quotations are often great places to jump off and have a discussion.

So Ali, please begin and go as long as you want on the low road to active inference.

Thanks.

Okay.

So, yes.

Chapter two, as the title suggests,

it's the construction of active inference theory

from the viewpoint of the low road active inference.

So it begins by providing this notion of perception as inference

because more often than not,

we usually think of perception as something just in a computational sense of sense

is a processing of the inputs,

or rather it's a raw processing of the inputs.

But here it shows that it can also be described as,

or even more accurately described as a kind of inference.

So it's not just a simple and raw processing of the inputs,

as my computational model,

or a computer analogy might suggest.

It's more like something that we predict,

and then we compare our inputs or the inputs we get from the stimuli

with our prediction,

and then try to somehow minimize that prediction error.

So I believe that this notion of perception as inference

is the most central notion of all the related theories of predictive coding,

predictive processing,

Bayesian brain hypothesis.

So up to now,

active inference is not very different from all the other theories.

It's basically a subset or a variant of those theories.

But then it progresses into distinguishing the active inference

from all the other theories and how it stands apart from the other ones.

So in section 2.1,

we began with some of the basics of probability theory,

namely the Bayes theorem in box 2.1,

and some simple examples about how Bayes theorem can be applied.

And again,

a very important page or important part of this section will be page 20,

which describes the concept of surprisal,

and specifically the statistical surprisal,

and how it differs from the phenomenological surprise or psychological surprise.

So,

and how we can formulate the statistical surprise.

Yes.

Yes.

Just to catch up here,

this example that's going to come back again and again

is a person who's guessing the object that's in their hand,

and it could be a frog or an apple,

and the object is going to jump or not.

And so that's used as a way to talk about the Bayesian updating

that forms the kernel of a variety of Bayesian brain-like models,

including active inference.

And then there are the presentation of the exact Bayes.

So in simple cases,

you can compute exactly what you want with Bayes theorem.

However,

it's a lot more effective on large data sets

to use certain approximations and heuristics

that we're going to be discussing.

And then,

Ali has highlighted that there are two concepts

that are very closely related to each other regarding surprise.

There's surprise by itself,

and then there's Bayesian surprise.

So please pick up there,

what are surprise and Bayesian surprise,

and why do they matter here?

Yes.

So,
well,
surprise,
I mean,
the regular surprise,
as we're all familiar with,
is something
more related to our sense of surprise
or psychological sense of surprise
of observing some unexpected phenomena
or unexpected behavior.

But Bayesian surprise,
or statistical surprise,
is,
of course,
closely related with
the psychological sense of surprise,
but a bit more rigorous
in which it's a way to compare
two probabilistic information
using callback Leibler divergence
and,
somehow,
getting the,
I mean,
how unexpected
that information
emerging from
callback divergence
and,
they call it
the surprisal.

So,
surprisal is,
in other words,
is a way to formulating
those unexpected
probabilistic information.

And,
it doesn't necessarily
align or
maps
onto perfectly
our psychological sense
of surprisal,
but,
as I said,
it's closely related
with that.

Yes.

Surprise.

By surprise,

please,

I mean,

for almost,

yes.

No,

you take it out.

I know,

I was just going to say that,

I was just going to say that,

by surprise,

almost,

in the rest,

all the other sections

and chapters of the book,

we basically mean

this Bayesian surprise

sense of the word.

And,

in order to distinguish

that,

with the regular

sense of surprise,

sometimes in the literature,

surprisal is used

to refer to

Bayesian surprise.

Yes.

So,

they're both measured

in information

theoretic units.

This is all happening

in information

geometric spaces.

The first concept

of surprise

is applied

to a given

single observations.

How surprising

is that one observation?

And so,

in that sense,

it's a lot like

the Z-score

of a data point

coming in

with respect

to a statistical distribution.

So,

it's like,

you have a height

distribution

in a classroom

and you measure

one person

and then you can say,

what is the Z-score

of that measurement?

Was it right

at the center

of the distribution

with a Z-score of zero
or were they
two standard deviations
higher with a Z-score
of two?

So,
it's kind of like that.
And that's why
there's a discussion
of probability distributions
and their support,
which is the X values
for which they're defined,
and surprise function
with the fancy I.

And that's going to be
a function
that helps you compute
how surprising
each observation
is given
a parameterization
of that distribution.

Whereas,
this Bayesian surprise
is more related
to learning.

it has to do
with how much
updating happens
between the prior
and the posterior
and that's before
and after
the observation.

So,
one could imagine
a surprising observation,

surprise concept one,
that either does
or doesn't update
the prior
into a very different
posterior.

So,
as Ali mentioned,
they are not exactly
the same,
but it's going to be
important to understand
how they're different
and that's worked out
again in the case
of the Apple
jumping.

Box 2.2
continues with
the discussion
of expectations.

Now,
expectation
in everyday parlance
might be
specifically referring
to something
in the future,
like I expect it
to rain tomorrow.

And,
in statistics,
when we talk
about the expectation,
we're talking
about the weighted
average
or the center

of gravity
of a distribution.
And that can be
in both
a discrete distribution,
at which point
you have a weighted
sum,
or a continuous
distribution,
in which case
it's an integral.

So,
you can have
an expectation
for the humidity
tomorrow,
and so that might
refer to the center
of gravity
at a t
equals plus one,
but just
taken alone,
expectation
means center
of gravity
of a statistical
distribution,
not anticipation.

section 2.3

is going
to describe
how some
of this
how,
this low
road that

we're on,
is going
to connect
to biological
inference.

There's
more discussion
of the generative
model and the
generative process,
and a little
bit of a hint
that the generative
model captures
aspects of the
generative process,
which is where
we see a lot
of the classical
cybernetic
theorems and
concepts like
requisite diversity,
good regulator
theorem,
and so on.

However,
the generative
model does
not have to
be exactly
isomorphic
with the
generative process.

For example,
the generative
process,
the temperature

in the room
might be a
continuous variable,
but then the
generative model
might be
discrete,
only modeling
integer-based
temperatures,
or might be
categorical,
like too hot,
just right,
and too cold.
So there's a
lot of
articulations
that can be
done because
of how
flexible and
interoperable
generative models
are with each
other.

Figure 2.2
is going to
expand upon
that chapter
one representation
of the
cybernetic
action perception
loop,
and we're
going to see
this more

in terms of
a generative
model and
generative process
articulation,
and there are
incoming,
speaking from
the perspective
of the
generative model,
the agent,
there are
incoming and
outgoing
dependencies in
this graph,
and this is a
little bit like a
schema,
more so than a
formal graph,
but also it is
like a Bayesian
graph,
where nodes are
variables and
edges are causal
relationships,
and so we have
the internal
states of the
model,
the external
states of the
generative process,
and then the
blanket states

that make
internal and
external states
conditionally
independent,
and again,
speaking from
the perspective
of the
agent,
although there
is a
symmetry,
we can talk
about the
incoming
sensory signals
happening from
the observations
handed from
the process
passed on to
the internal
states of the
generative model,
and then the
outgoing actions
that are selected
that then can
influence the
hidden state of
the world,
for example,
going and
turning on the
heater to
increase the
temperature in

the room,
and so this
action-perception
loop or
particular partition
is going to get
explored in a lot
more detail in
the coming
sections.
Previous
section 2.3
was on
perception as
inference,
and now
action as
inference is
going to be
discussed,
and that is
where they say
the discussion
to this point
is common to
all Bayesian
brain theories,
we now introduce
the simple but
fundamental advance
offered by
active inference,
which is the
extension of
this inferential
perspective to
consideration of
action as

inference.

Perception and
action cooperate
to realize a
single objective.

Section 2.5 is
about minimizing
the discrepancy
between the
model and the
world.

We already
described that
there's two ways
for this to
happen, change
your mind and
change the
world.

That's how the
discrepancy can
be reduced or
managed.

We see a
variant of the
action perception
loop where the
agent is making
perceptions,
predictive
models of the
world that
through perception
are being
juxtaposed with
observations handed
from the
generative process,

the world, and
a discrepancy is
realized, some
non-zero
discrepancy.

And then here
are those two
paths to
minimize free
energy.

Change beliefs
by perception
and learning or
change the world
through action.

Section 2.6 is
going to discuss
how the exact
Bayesian approach
described earlier
is absolutely
spot on if you
have infinite
computing resources.

However, we're
often interested
in rapid or
large data
sets where we
want to be able
to get approximate
Bayesian
computation or
probably approximately
correct computation
in a vastly
accelerated fashion.

And so that is

going to be
approached using
what's called
variational Bayesian
inference that's
unpacked in
chapter 4.

But it suffices to
say that
variational Bayesian
inference implies
substituting two
intractable
quantities,
the posterior
probability and
log model evidence
with two quantities
that approximate
them but can be
computed efficiently,
the approximate
posterior Q and
a variational free
energy.

So it transforms
an intractable
estimation problem
into a highly
tractable
optimization problem.

Pick up from
there, Ali.

Yes, sorry,
I just wanted
to point out
a couple of
things, especially

about section
2.3, because
I believe it
is one of the
crucial sections
of this chapter
and in fact
the whole book
because it
provides some
of the
justifications of
using Bayesian
inference as
opposed to
some other
mathematical
techniques such
maximum likelihood
estimation.

But more
important than
that, I think
it's this
concept of
optimality which
comes into play
in almost
everywhere in
the literature
and of course
in this book.

So there are
actually two
notions of
optimality
which is not
discussed in

detail here,
namely Bayesian
optimality and
Jane's optimality.
But in some
of the recent
papers on
Bayesian
mechanics,
Dalton
Sack,
DeVade,
L-Max,
Rampstead,
and others have
shown that
those two
concepts of
optimality are
actually congruent
with each
other.
So that's
one of the
reasons
that
the duality
between
FEP,
free energy
principle,
and constrained
maximum
entropy
principle
can be,
I mean,
it's one of

the justifications
for
providing that
dual formalism
between those
two.

So,
but another
point I wanted
to mention
here is,
because I've
seen that
using the
word hidden
state can
be a bit
confusing for
some people,
because when
we observe
something as
an observation,
obviously it
is not
quote-unquote
hidden,
right?

So what
hidden state
here refers
to is actually
the hidden
cause of that
observation.

So it's not
that the
observation

itself is
hidden from
the observation
or it's
unobserved.
so that
might be a
bit confusing
if we
don't take
into consideration
the exact
meaning of
the hidden
state or
latent
state here
and in the
rest of the
literature.

So,
continuing from
section 2.6,
here we see
one of the
two central
equations of
active inference,
equations,
which is
equation 2.5
for
variational
free energy.

So,
it's,
as I said,
I mean,

understanding
this equation
and how
each line
of its
formulation
represents
in terms
of the,
I mean,
trade-off
between energy
entropy
or complexity
inaccuracy
or divergence
and evidence
is key to
understanding
almost everything
in the rest
of the book
and in many
other literature
on active
inference.

So,
this is
the perceptual
part of
active inference.

So,
variation
free energy
is a
parameter
that
is a

notion
that
parameterizes
the
surprisal
of our
perceptual
information
we have
about the
external
states.

And then
we'll see
in the
next section
the
related
and almost
symmetrical
formulation
to
variational
free energy
namely
expected
free energy
which is
basically
the
action
part of
the
active
inference.
So,
we can
see how

those two
can somehow
be seen
as a
kind of
unified
formalism
but described
in
alternate
expressions.

So,
but
another thing
about
equation

2.5
is
it
may be
a bit

I don't
know
daunting
to
see

all the
relations
between
those three
lines of
equation
and how
we can
get from
one to
the other.

So,

there are
some
supplementary
materials
that we
have developed
in the
past
weeks
which I
think
can help
in
clarifying
how the
derivations
of these
three lines
of equation
be done.

So,
I hope
they would
be
clarifying
in,
I mean,
and help to
understand how
those three lines
relate to
each other.
so,
but the key
point here
is
to understand
that

variation of
free energy
is not
something
absolute,
but it's just
an upper
bound
for the
minimization.

So,
as Daniel
just mentioned,
it's
untractable
to have
an absolute
amount
for the
surprisal
to be
minimized.

So,
we need
to have
an upper
bound
in order
to make
that
more
tractable
because
by
Jane's
inequality
as
we'll

saw
in
chapter
four,
I think,
we can
see
that
how
the
upper
bound
of a
surprisal
necessarily
provides
a condition
for
minimizing
the
precise
or
the
exact
free
energy.
And
that's
the key
insight
of
equation
2.5
or
the
notion
of
variation

of
free
energy
which
is
to
provide
this
upper
bound
instead
of
the
exact
amount
of
surprisal
to be
minimized.

Yeah,
what I'll
add there
is
if you
knew
exactly
how
well
you
should
be
surprised
by
a
given
data
point
y,

then
you
would
have
had
the
optimal
model.
However,
that is
not
tractable,
and so
by making
a quantity
that's
always
higher
or an
upper
bound,
the
variational
free
energy
 f ,
which is
a
function
of
broadly
 q ,
our
beliefs,
our
variational
beliefs,
which are

built in
a way
that makes
them
very
compositional,
very
optimizable,
very
interpretable,
and data.

And we
can reduce
the
divergence,
here the
KL
divergence
with a
double
line,
between
 q ,
our
beliefs,
and p ,
the kind
of
actuality
of it,
and if
we can
reduce
this
divergence,
in other
words,
minimize

the free
energy,
then we
will come
closer and
closer to
the true
surprise
function,
and do
that in
a
tractable,
incrementally
optimizable
way.

So,
equation
2.5
is going
to be
the
variational
free
energy,
different
ways that
it can
be
represented
as it
takes
in
data
and
beliefs
about
the

world,
q,
and then
this is
going to
be
expanded
into
the
future
to
include
action
with
g,
the
expected
free
energy.
Now,
there's a
lot more
that we
can say
about this.
There's a
lot of
technicalities
to go
into,
but broadly
notice that
g,
the expected
free
energy,
is a
functional

of
policy,
pi,
because it's
only being
evaluated
to select
amongst
different
action
outcomes,
pi,
and another
important
difference is
that it's
going to
be describing
sensory
outcomes
that haven't
yet
happened,
a sort
of what
would I
perceive
if I
did
A or
what if
I did
B,
and it's
that kind
of comparison
that allows
the expected

free energy
functional
here to
be used
in action
selection
or policy
selection
as inference,
planning
as inference.
That section
is expanded
upon,
and in
figure 2.6,
we see a
very nice
representation
of the
expected
free energy
equation,
and then
how when
certain aspects
of this
equation
or situation
are zeroed
out,
we get
certain
other
familiar
cases.
For example,
where there's

no epistemic
value,
there's no
information
to learn,
then you
get
ruthless
expected
utility
theory.

And
conversely,
where there
is no
pragmatic
value to
extract,
so all
outcomes
are equally
valid
or
preferable,
then you
get things
like Infomax
principle
and optimal
Bayesian
design.

And then
everything in
between is
the space
that we're
interested in.
And so

this
figure 2.6
shows that
the expected
free energy
functional
can be seen
as like a
generalization
of a lot
of other
settings
related to
perception
and action
and planning
amidst
uncertainty.

And
section
2.9
closes
the low
road.

They took
us all
the way
to active
inference
from Bayes
theorem
through the
generative
model
on to
active
inference
and

clarifies
these two
notions
of
variational
free
energy.
That's
the
real-time
perceptual
unfolding
evidence
lower bound
unsurprisal
tractable
optimizable

and so
on.
And
F
and
G
the
expected
free
energy
which
is able
to do
planning
as
inference
or policy
selection
as
inference.

Expected
free
energy
is
fundamentally
prospective
and that
enables
counterfactual
cognition.

Section

2.10

summarizes
active
inference
is a
theory
of how
living
artifacts
underwrite
their existence
by minimizing
surprise
or
attractable
proxy
to
surprise
variational
free
energy
via
perception
and
action.

And
they

motivate
that
from
a
first
principles
base
theorem
starting
place.
Any
closing
thoughts
on
chapter
2,
Ali?
I would
just
humbly
suggest
for the
people
who
want to
go through
this
chapter
to
try
their
best
to really
understand
and
specifically
what
equation

2.5

and

2.6

represents

and how

I mean

how to

use

those

equations

to

describe

different

situations

with

some

missing

elements

as well

because

I believe

those

sections

and

particularly

those

equations

are

absolutely

essential

for

understanding

everything

active

inference

related

both

in the

rest of
this
book
and
in
almost
all.
thanks
and
just
a
last
thought
I'll
give
on
chapter
two
is
this
is
exactly
the
work
that
we
do
in
the
active
inference
textbook
group
we
really
welcome
all
backgrounds

every
single
question
and
uncertainty
you
have
is
beautiful
we
have
a lot
of
resources
that
Ali
and
others
make
to
make
the
math
approachable
and
rigorous
and
natural
language
descriptions
of the
equations
!
that
gives
meaning
to
our

usage
and
fluency
of
the
active
inference
ontology
which
Ali
and I
are
speaking
right
now
we're
not
just
saying
surprise
is
related
to
this
because
we
felt
it
that
way
there
is
an
underpinning
and
it
is
a

really
interesting
life's
work
to
explore
it
but
we're
finding
ways
to
communicate
it
and
learn
and
teach
it
better
and
better
every
time
so
that's
chapter
two